Juliette Bruce's Future Plans

My research interests lie in algebraic geometry and commutative algebra, which are two fields that study the same objects from opposite sides. A parabola in the plane can be described geometrically as DINIS, or algebraically, as the solutions to the equation $y = x^2$. Each of these descriptions provides differing perspectives on the same questions. For example, the face that a straight line intersect a parabola in at mot two points (geometry) is precisely because a quadratic equation has at most two solutions (algebra). This dictionary between algebra and geometry is at the heart of my field, and motivates much of my research. Questions about shapes (lines, parabolas, curves, surface, etc.) becomes questions about polynomial equations and vice versa.

A running theme throughout much of my research is that the algebra of the polynomial equations defining a shape capture its geometry in subtle ways. For example, the equations defining a shape are rarely independent. Instead, the equations normally satisfy a separate set of equations (i.e. relations) among themselves, called syzygies (from the Greek for "yoked together"). The equations defining these relations then satisfy their own relations, and so on. Amazingly, Hilbert showed that this cascade of relations eventually terminates, resulting in a finite list of relations (i.e. syzygies) that capture some of our shape.

Over the next several years, my research program seeks to carry this dictionary into new settings in three closely connected directions. The first two directions seek to extend the classical story of syzygies and a measure of geometric complexity called Castelunuov–Mumford regularity to settings that have long been out of reach: curves with extra symmetries (stacky curves) and shapes built out of polyhedra (toric varieties). The third direction seeks to a third language to the algebra–geometry dictionary discussed above: combinatorics. A running theme running through all three of these directions is tackling longstanding hard geometric questions by translating them into algebra and combinatorics, which are more amenable to ADDD

Syzygies of Stacky Curves. Curves are the among the most classical objects of studied in algebraic geometry and mathematics more broadly. A forty-year old program begun by Mark Green, and culminating in celebrated work of Claire Voisin, showed that the syzygies of a curve, in its most natural embedding, can be used to detect how special the curve is. The precise conjecture, known as Green's conjecture, is one of the jewels of the field.

Stacky curves are curves where extra symmetries are encoded by allowing a finite number of special points on the curve FINISH. Such spaces re not exotic and arise naturally, for example, the spaces that organize modular forms – modular forms are highly symmetric functions at the heart of number theory and central to the proof of Fermat's Last Theorem – are stacky curves. Despite this prevalence, the classical theory of syzygies does not apply to them as they require variables whose degrees are not one. The equations of stacky curves were only recently described by Voight and Zureick-Borwn, and their syzygies are uncharted.

With M. Banks and P. Wei, I am building this missing theory. We have already found the overall shape of FINISH. This has led us to a stacky Green's conjecture: the special points shift the classical answer in a predictable way (exactly the number of extra equations they require). In the next several years I hope to develop the tools needed to prove this conjecture. I plan to begin by considering the simplest possible stacky curve where there is is only one spacial point.

Multigraed Regularity. Classically, the Castelunuov–Mumford regularity of an algebraic variety is a single number that measures the complexity of the shape. It turns out to be central to many FIHSIN. This prevelance is underlined by a celebrated 1984 theorem of Eisenbud and Goto shows that three very measure of complexity that are on their face very different (thourgh geometyr, through syzygies, and through the degrees of the equations) all give the same answer: Castelunuov–Mumford regularity.

However, many important shapes, such as the stacly curves in the previous sectioj, do not live in ordinary (projective) space, but instead in spaces built from gluing projective spaces toegther in ways determiend by polyhedra. In these more general ambient spaces Maclagan and Smith gave the correct generalziation of Castelunuov–Mumford regularity that they call multigraded regularity. In this theory regularity is no longer a single number, but instead is n ininifte region of lattice points bounded by a staircase type shape that can be remarkably intricate. For instance with my collaborators I showed that even if one is only looking at the simplest possible geometric shape, a point in a toric surface, the staircase can have infinitely many corners, a pathology the no classical analog.

My plans aim to understand and tame this behavior. First, I have conjectures that the infinite staircases are an artifact of measuring complexity with respect to a cone that does not capture all of the geometry. In particular, showing that there is a precise duality between the regularity staircase and the set of curves on a toric variety. If sucessful this would not only improve our theoretical understanding of multigraded regularity, but also provide the first computational algorithm to fully computing regularity. Second, I will prove a multigraded analog of a theorem of Gruson, Lazarsfeld, and Peskine which bounded the Castelunuov–Mumford regularity of a curve purely in terms of the number of times it crosses a general line (e.g., for a parabola 2).

Finally, and more speculatively, I will explore whether regularity is asking the right question, but in the wrong algebraic setting. Recent, work has shown that the algebra better captures the geometry of toric varieties when one studies a toric variety and all of its close relatives all at one. Inspired by this I am exploring whether a similar phenomena is true for regularity. In particular, I would like to give a new definition of regularity using a recently invented algebraic tool called the Cox category. If sucessful this theory would resolve Oda's conjecture, a long standing open probelm in toric and polyhedral geometry.

Topical Geometry of Moduli Spaces. A *moduli space* is a space whose points themselves represents shapes and nearby points represent similar shapes. Despite this, many of the basic questions about the topology of such spaces remain unknown. For example, Gurshevsky asked where the moduli space of abelian varieties has nonzero odd cohomology classes – roughly the number of holes a shape has in each dimension.

Tropical geometry offers a way in; it replaces the complicated objects with their combinatorial shadows, curves are replaced with graphs, abelian varieties are replaced with polytopes, and so on. Building on breakthrough work of Chan, Galatius, and Payne for curves, my collaborators and I adapted these methods to abelian varieties. This provided numerous new cohomology classes, and positively answered Gurshevsky question (*Geometry & Topology*, 2024). Since this moduli space is closely related to the group of symmetries of integer matrices, our relates also give new information about these groups.

A central aspect of my research program over the next several years is to develop these methods into general tools. Work of Deligne shows that the cohomology of a moduli space naturally splits into layers, called weights. The work of Chan–Galatius–Payne and myself and collaborators shows how tropical geometry captures exactly one of these layers, namely the weight-zero layer, which it turns out is entirely determined by the combinatorial shadow of the moduli space. My plans for the ufutre is to push this futher in three directions. First, I am to enrich the tropical spaces we study so that they themseves are moduli spaces, this requires using new notations of spaces called cone stacks. Together with Raluca Vlad we have proven prelimary results suggesting this approach will be sucessful. tSecond, I hope to give a precise geometric comparision showing how the weight-zero layer can be described explicitly from the tropical moduli space. Third, I will push into the deeper, more mysterious, layers of cohomology by studying the weight-two cohoomology of the moduli space of abelian varieties. Doing this requires enriching the classical combinatorics of tropical geometry so that the combinatorial shadows carry slighly more geometric infomration.

Computation, Open-Source Software, & Community Center AI. Computing examples to develop conjectures and survery the landscape of a problem is a central driver of much of my theoretical work. For example, using high-performance high-throughoutput computing my collaborators and I computed the syzygies of hundreds of new surfaces byeond previous reaach (*Expiremental Mathematics / Journal of Algebra* ADD). Further, I have co-authored four software packages for the open-source computer algebra system *Macauly2*. I plan to coninue building upon this work in three ways. First, since no software exsits for computing with stcky curves, M. Banks, J. Cobb, and I are develop `toricStacks`, a packge for *Macualay2*. Second, I am building a searchable database of thousands of expert-chosen mathematical objects and examples (ideals, rings, modules, etc.) that are hidden in the source code and documentation of *Macauly2*. By making these examples easily accesibly this databes will make testing conjecture, benchmarking software, and developing community centered AI easier. This, with J. Voight I am planning a workshop focused on the future of mathematical software in the age of AI.